SCORE: TOKENIZATION-AWARE TRANSFORMER FOR SYMBOLIC MUSIC GENERATION WITH EXPRESSIONS

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ABSTRACT

Ever since the electrification of guitar, there has been an explosion in rhythmic complexity, tone experimentation and playing techniques. In this paper, a novel tokenisation method is proposed, which encapsulates the various guitar techniques and a sequence level attention component for the transformer architecture, which captures segmentation between musical phrases. The given approach enables efficient batching of variable length musical phrases while preserving their contextual coherence.

1 INTRODUCTION

A *riff* is a short, repeated motif or figure in the melody or accompaniment of a musical composition (Randel, 1986). This type of composition is particularly prevalent in rock, metal, blues, funk and jazz. It is often the main feature that gives character and structure to a song. Developments in Transformers architecture have inspired many to generate sequences by treating musical notes as symbolic language (Huang et al., 2018) (Huang & Yang, 2020a) (Wu & Yang, 2020) (Ens & Pasquier, 2020) (Muhamed et al., 2021) (Yu et al., 2022). (Dong et al., 2023) (Ferreira et al., 2023) (Agostinelli et al., 2023) (Copet et al., 2024) (Cui et al., 2025). Generating modern music demands capturing not only note patterns but also the associated expressions.

A widely adopted method for representing musical notes digitally is the MIDI format (Craig, 2025), which encodes pitch, duration, and dynamics. Using such a format for musical expression has limitations like resolution and bandwidth constraints (Shamoon, 2024) when compared to richer representations like audio recordings and detailed scores. Modern guitar riffs frequently employ intense and abrupt expressive techniques, which contribute significantly to its overall identity (Vallejo, 2021). To accurately represent the nuanced expressive elements of modern music, a score based format is preferable over simpler encodings like MIDI. The proposed method requires a tokenisation scheme that represents notes, beats, beat types and their associated accents, necessitating a dataset that includes expressive performance details.

In the context of symbolic music, notes are represented as sequences of discrete tokens. This requires a tokenisation scheme which directly affects how effectively a model can learn and generate coherent musical structures. Most existing approaches rely on methods similar to text processing such as BPE (Byte Pair Encoding) (Sennrich et al., 2016), WordPiece (Schuster & Nakajima, 2012) and SentencePiece (Kudo & Richardson, 2018). Although subword tokenization improves the structure (Kumar & Sarmento, 2023) (Fradet et al., 2023), it becomes limited when dealing with the expressive element combinations of modern music. Recent studies (Gardín, 2024) have explored modeling chords based on scale degree, tonality and octave, and notes encoded relative to the current chord. This approach, however, is unable to produce every possible chord, particularly when dissonance is involved, which is becoming increasingly popular in modern metal. Most prior studies use absolute pitch values (Suresh et al., 2020), some (Kermarec et al., 2022) investigate the use of pitch intervals. This introduces an additional processing step, increasing inference time. Another common strategy treats each unique chord as a separate token, resulting in a large vocabulary that may include thousands of tokens representing all possible chords in the dataset (Dalmazzo et al., 2024b) (Dalmazzo

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Table 1: Beat type. d = 7 for dotted note, 0 otherwise.

b_t	**Value**
Triplet(3)	1 + d
Quintuplet(5)	2 + d
Sextuplet(6)	3 + d
Septuplet(7)	4 + d
Nonuplet(9)	5 + d
Undecuplet(11)	6 + d

Table 2: Beats.

b_b	**Value**
1 / 64	0
1 / 32	1
1 / 16	2
1 / 8	3
1 / 4	4
1 / 2	5
1	6

et al., 2024a). Chen et al. (2020) proposes a pre-defined set of rhythmic groove patterns. This limits the model's ability to generate novel rhythms.

Despite the growing body of work on symbolic music generation, there is little to no prior work specifically addressing expressive guitar music that captures techniques such as bends, slides, and chords in a tokenized representation. To address these limitations, a tokenisation scheme is proposed, that captures single and multiple musical notes efficiently and their associated accentuated elements. Furthermore, a sequence level attention mechanism is introduced to capture the segmentation between different musical phrases, facilitating batching of variable length phrases while maintaining musical context. The modified architecture is trained and evaluated on a dataset (Begari, 2025a) that includes guitar specific techniques.

While direct comparison with existing methods is not feasible due to differences in model compatibility and expressive feature representation, the proposed method is evaluated using quantitative metrics. The model is trained to minimize KL divergence across pitch, beat, beat type, and expression similarity. The source code of this work is available at `https://github.com/DjentleViBe/SC0RE`.

2 Methods

2.1 Tokenisation

2.1.1 Note representation

In place of absolute pitch indices, both fret and string numbers are employed to represent notes. This approach allows for a more compact representation of notes, as the same pitch can be played on different strings and frets. This is particularly relevant to guitar music, where notes can be produced at multiple fretboard positions, often resembling chord shapes. The model focuses on learning the relative motion of notes across frets, emphasizing positional changes rather than absolute pitch representation. This provides flexibility to accommodate alternative tunings (Weissman, 2006). The formula for note encodings for fretted string instrument is given by:

$$n_e = n_i + 2b_t \cdot n_s \cdot n_{ps} + 2b_b \cdot n_s \cdot n_{ps} \tag{1}$$

where, n_s = number of strings, n_{ps} = number of pitch steps per string.

$$n_i = f_n + s_n \cdot n_{ps} + n_s \cdot n_{ps} \cdot n_{val} \tag{2}$$

where, f_n = fret number, s_n = string number and n_{val} = 1 for palm mute, 0 otherwise; b_t and b_b are defined in 1 and 2 respectively.

2.1.2 Accent representation

A description of expressive elements is provided in (Begari, 2025b). An accent token is appended to the note token from Section 2.1.1 to indicate the presence of an expressive element. For chords, the accent token is added before the next consecutive note token. A C major chord as shown in Figure 1 comprises of 3 notes - root note (C) played on the fret 2, 5th string , major third (G) on fret 1, 4th string and perfect fifth (E) on fret 2, 1st string (Alexander, 2019). The tokenization for the chord is

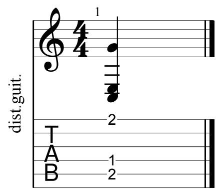

Figure 1: C major chord shape on a 6 string. The diagram illustrates muscial notation along with guitar tablature.

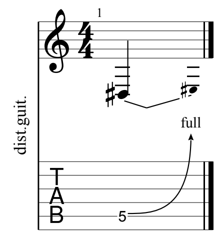

Figure 2: Musical notation and guitar tablature showing full note bend on fret 5, string 5.

as follows: `15413, 27051, 15389, 27051, 15321, 27051`. Here, `27051` is the chord accent token, BARRE_NOTE[A]. Similarly, for a bend note depicted in Figure 2, the tokenization is: `15461, 27052`. Here, `27052` is the bend accent token, BEND_NOTE_1[A].

2.1.3 Phrases

A music bar (also called a measure) is a fundamental unit in musical notation that divides music into segments of equal time, defined by the time signature (Miller, 2016). Each measure is separated by a BAR[A] token.

2.2 Embedding

A bar consists of a fixed number of beats, composed of musical notes that were tokenised as described in Section 2.1. Each bar is treated like a sentence. It has been demonstrated that the transformer equipped with relative attention is very well-suited for generative modelling of symbolic music (Huang et al., 2018) (Zhou et al., 2023) (Li et al., 2024). The chunking strategy used to carry over context between sequences during training follows the approach introduced in Yang et al. (2019), which builds upon Transformer-XL's (Dai et al., 2019b) segment-level recurrence mechanism. This method enables the model to capture longer-range dependencies by reusing hidden states from previous token chunks across segments. An example of a riff is shown in is given in Figure 3.

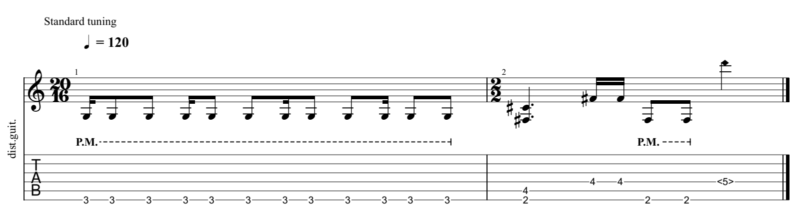

Figure 3: Example guitar riff showing a sequence of notes in combination with accents like pinch harmonic, barre chord and palm mute.

An example of 3 sequences with 16 tokens per sequence embedding is as follows:

```
[27049, 7985, 11849, 11849, 7985, 11849, 11849, 7985, 11849,
11849, 7985, 11849, 11849, 27049, 17485, 17506]
[11849, 11849,  7985, 11849, 11849, 27049, 17485, 17506, 27051,
7802, 7802, 11848, 11848, 15530, 27074, 27049]
[27051,  7802,  7802, 11848, 11848, 15530, 27074, 27049, 27075, 0,
0, 0, 0, 0, 0, 0]
```

The first two sequences contain musical notes, while the third sequence is padded with zeros to maintain a consistent length of 16 tokens. This padding ensures that all input sequences are of uniform length, while still allowing the flexibility to handle variable-length musical phrases.

2.3 Training

2.3.1 Dataset

The guitar tracks were obtained from Begari (2025a). The dataset consists of files in .gp5 (Abakumov, 2025) format in standard tuning. From the full set, 6208 sequences were randomly sampled to enable efficient training on standard desktop hardware. The dataset was split into training, validation, and test sets using the common 80/10/10 ratio (Goodfellow et al., 2016). For inference, the first 10 notes of each measure from the training set were selected as prompts for sequence generation and evaluation. Relavant statistics about the dataset used for training and evaluation are provided in Begari (2025b).

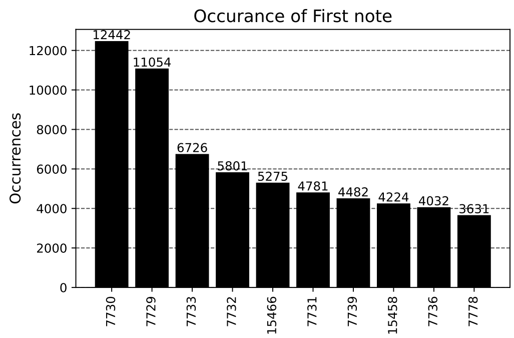

Figure 4: Distribution of starting notes in training data for musical measures, used to select initial tokens at inference time for more consistent generation.

2.3.2 Model

The Transformer is built from a decoder-only stack with masked self attention as described in Huang et al. (2018). Although, segment-level recurrence mechanism is captured in Transformer-XL (Dai et al., 2019b), its memory is limited to one previous segment at a time. To model dependencies across a whole batch, the gaussian distributions of the sequences are learned through a Multi layer Perceptron layer. If the input is a batch of sequences defined by:

$$X \in \mathbb{R}^{B \times T \times d} \tag{3}$$

where:

- B: batch size (number of sequences)
- T: number of tokens per sequence
- d: embedding dimension
- $X_i \in \mathbb{R}^{T \times d}$: the i-th sequence in the batch

Sequence level mean pooling is performed by:

$$\bar{x}_i = \frac{1}{T} \sum_{t=1}^{T} X_{i,t} \quad \in \mathbb{R}^d \tag{4}$$

Gaussian parameters are then computed with a Multi Layer Perceptron. Let $f : \mathbb{R}^d \to \mathbb{R}^{2d}$ be a learned MLP. Then:

$$[\mu_i, \log \sigma_i] = f(\bar{x}_i) \quad \text{with} \quad \mu_i, \log \sigma_i \in \mathbb{R}^d \tag{5}$$

Ensure positivity of standard deviation:

$$\sigma_i = \exp(\log \sigma_i) + \varepsilon \quad \text{where } \varepsilon > 0 \tag{6}$$

Each sequence is then represented as a diagonal multivariate Gaussian:

$$\mathcal{N}_i = \mathcal{N}(\mu_i, \mathrm{diag}(\sigma_i^2)) \tag{7}$$

The pairwise KL divergence between each pair of Gaussians (Zhang et al., 2023):

$$\mathrm{KL}(\mathcal{N}_i \parallel \mathcal{N}_j) = \frac{1}{2}\sum_{k=1}^{d}\left[\log\left(\frac{\sigma_{j,k}^2}{\sigma_{i,k}^2}\right) + \frac{\sigma_{i,k}^2}{\sigma_{j,k}^2} + \frac{(\mu_{i,k}-\mu_{j,k})^2}{\sigma_{j,k}^2} - 1\right] \tag{8}$$

This produces a KL matrix:

$$\text{KL\_matrix} \in \mathbb{R}^{B\times B}, \quad [\text{KL\_matrix}]_{i,j} = \mathrm{KL}(\mathcal{N}_i \parallel \mathcal{N}_j) \tag{9}$$

A simpler version of the attention matrix from Dai et al. (2019a) is used from which the KL divergence-based sequence bias is subtracted:

$$\mathrm{RA}_{\mathrm{KL}} = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + QR^\top - \alpha\cdot \mathrm{KL}_{\text{bias}}\right) \tag{10}$$

A weighted sum over relative positional value contributions V_{rel} is then added:

$$\text{Output} = \mathrm{RA}_{\mathrm{KL}} \times V + \mathrm{RA}_{\mathrm{KL}} \times V_{\text{rel}} \tag{11}$$

2.3.3 Loss

The model's loss is composed of three components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_{\text{rep}} + \delta\mathcal{L}_{\text{seq}} \tag{12}$$

Cross Entropy loss ($\mathcal{L}_{\text{CE}}$) Cross-entropy loss is the standard choice for training deep neural networks and logistic regression models for both binary and multi-class classification tasks (ml glossary, 2017).

Repetition loss ($\mathcal{L}_{\text{rep}}$) Repetition helps organise musical ideas, making pieces more coherent and memorable. It gives listeners reference points and expectations, aiding in the recognition and recall of themes or motifs (Hu et al., 2022). Composers have long used stochastic (probability-based) methods to inject randomness into repeated motifs, sometimes even employing quantum randomness for truly unpredictable results (Yamada et al., 2021). This demands careful consideration, as it relies on the listener's inherent ability to recognise patterns and their natural inclination toward novelty. Since it is already known that accent tokens [A] cannot repeat, it is penalised through repetition loss, making them less likely to be selected again in subsequent steps. The concept of the repetition penalty introduced in Keskar et al. (2019) uses penalised sampling at inference time. The proposed loss is a logits regulizer which is fully differentiable and does not require autoregressive sampling during training, making it compatible with teacher-forced optimization in decoder-only architectures. For batch element $b \in \{1,\dots,B\}$ and timestep $t \in \{1,\dots,T\}$, let $\mathbf{z}_{b,t} \in \mathbb{R}^V$ be the output logits of the decoder.

The softmax distribution with temperature $\tau > 0$ is defined as:

$$P_{b,t}(v) = \frac{\exp(z_{b,t,v}/\tau)}{\sum_{u=1}^{V}\exp(z_{b,t,u}/\tau)}, \tag{13}$$

where $v \in \{1,\dots,V\}$.

Let $\mathcal{V}_p \subseteq \{1,\dots,V\}$ denote the set of penalized tokens. The total probability mass assigned to penalized tokens at timestep t is defined as:

$$S_{b,t} = \sum_{v\in\mathcal{V}_p} P_{b,t}(v). \tag{14}$$

The logits-based penalized token adjacency loss is then defined as:

$$\mathcal{L}_{\text{adj}} = \frac{1}{B(T-1)}\sum_{b=1}^{B}\sum_{t=2}^{T} S_{b,t-1}\cdot S_{b,t}. \tag{15}$$

This loss penalizes the model for assigning high probability mass to penalized tokens at consecutive timesteps, including both identical token repetition and cross-token adjacency within $\mathcal{V}_p$.

Sequence loss ($\mathcal{L}_{\text{seq}}$) A sequence memory module is defined that maintains embeddings of sequences. Let N denote the number of sequences and d_{model} the embedding dimension. For a sequence ID $i \in \{1, \ldots, N\}$, the embedding is given by:

$$\mathbf{s}_i = \text{Embedding}(i) \in \mathbb{R}^{d_{\text{model}}}. \tag{16}$$

Given hidden states from a transformer model

$$\mathbf{H} \in \mathbb{R}^{B \times H \times T \times D},$$

where B is the batch size, H is the number of hidden heads, T the sequence length, and D the hidden dimension per head, we permute and flatten the hidden states:

$$\mathbf{H}_{\text{perm}} = \mathbf{H}.permute(0, 2, 1, 3) \in \mathbb{R}^{B \times T \times H \times D}, \tag{17}$$

$$\mathbf{H}_{\text{flat}} = \text{reshape}(\mathbf{H}_{\text{perm}}) \in \mathbb{R}^{B \times T \times (H \cdot D)}. \tag{18}$$

The sequence-level embedding is then obtained by averaging over time steps and adding the sequence memory embedding:

$$\mathbf{h}_{\text{seq},i}^{\text{final}} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{H}_{\text{flat},i,t,:} + \mathbf{s}_i, \tag{19}$$

for $i = 1, \ldots, B$. To enforce consistency between the current sequence embedding and the learned sequence memory, the mean squared error (MSE) loss is defined as:

$$\mathcal{L}_{\text{seq}} = \lambda \cdot \frac{1}{B} \sum_{i=1}^{B} \left\| \mathbf{h}_{\text{seq},i}^{\text{final}} - \mathbf{s}_i \right\|_2^2, \tag{20}$$

where λ is a sequence penalty weight hyperparameter.

This loss encourages the model to maintain a representation close to the learned sequence memory, improving sequence-level consistency across batches.

3 Results

The idea of context carryover has been well established in Dauphin et al. (2016) Wei et al. (2021) Chen et al. (2025) Rae et al. (2019) Beltagy et al. (2020) Zaheer et al. (2021). During inference, the model carries over context from previous sequences by using the last token of the preceding measure as the first token for the subsequent measure, following the approach described in Huang & Yang (2020b). To compare the quality of predicted tokens (P) from true distributions (Q), relative entropy is calculated using Kullback-Leibler divergence, following the methodology in Seo et al. (2016) Zhang et al. (2025) Spiliopoulou & Storkey (2011):

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in X} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \tag{21}$$

3.1 Grid search

To determine the optimal configuration, a grid search was performed over a predefined set of hyperparameter values. The search included combinations of key parameters such as learning rate, sequence length, model dimensionality, Each configuration was trained for 10 epochs, and performance was evaluated on a validation set using loss. The best-performing set of hyperparameters was selected based on the lowest validation score.

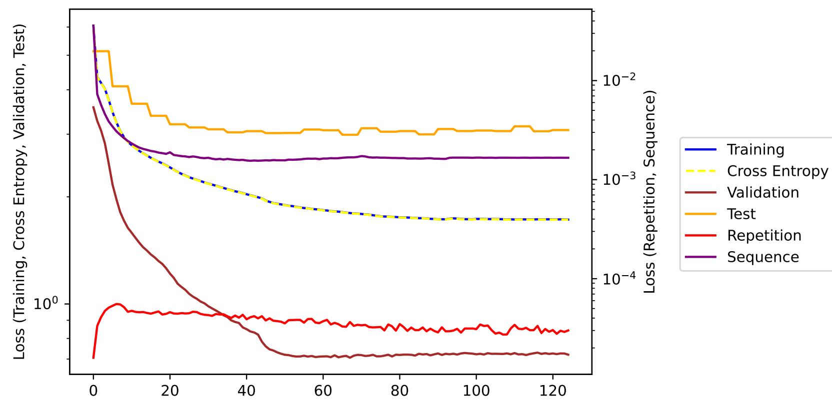

Figure 5: Loss plots over 125 epochs for a median of 3 independent training runs for hyper-parameter values listed in Table 4. Test loss is computed every 5 epochs.

3.2 KL Divergence Evaluation

To quantitatively assess the model's generative quality, the predicted token distributions were compared with the true distributions using KL divergence. Figure 6 visualizes the results. Given that the model is autoregressive and expects an initial token, selecting the start token based on the distribution of starting notes of musical measures in the training data yields more consistent generation. The results were averaged over 10 generated phrases per prompt.

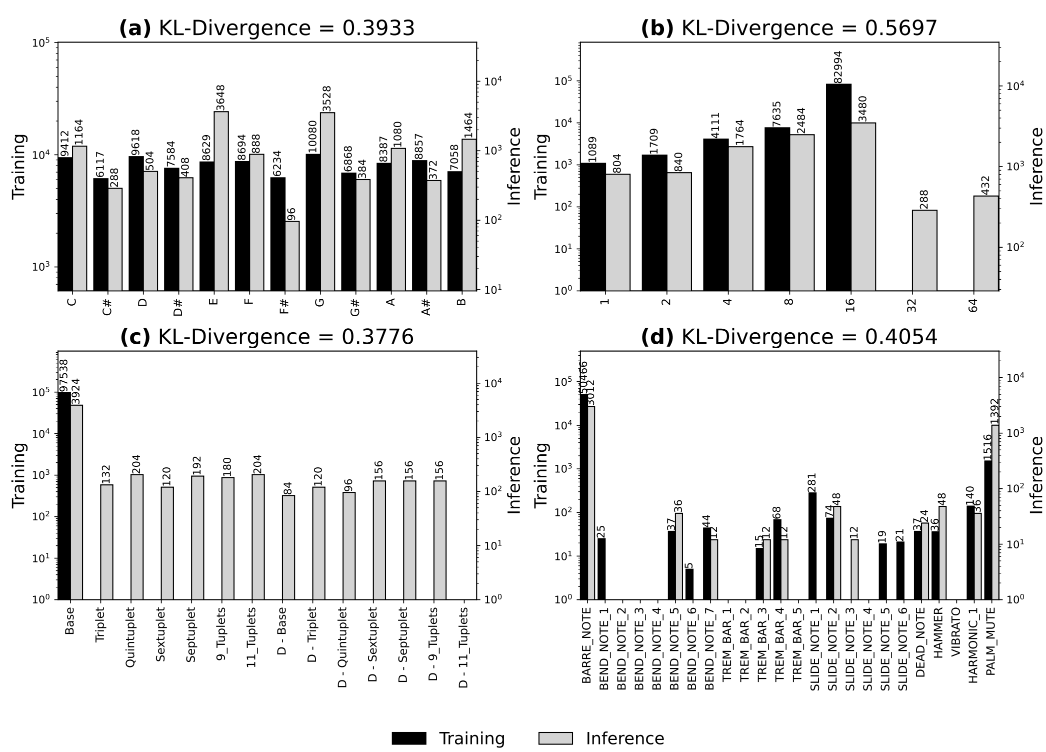

Figure 6: KL divergence between inference and training distributions **(a)** Musical notes, **(b)** Beats, **(c)** Beat Types, **(d)** Accents, indicating prediction by the model.

4 CONCLUSIONS

The proposed tokenisation scheme efficiently incorporates expressive elements without needing an explicit representation like chord vocabulary. The sequence-level attention mechanism captures phrase segmentation, enabling variable-length batching while preserving musical coherence. The model demonstrates strong performance in generating guitar riffs with expressive details, as evidenced by low KL divergence scores across pitch, beat, and beat type distributions. By incorporating a repetition penalty loss, the model naturally reduces redundancy without requiring explicit inference constraints. Future work could explore extending the approach to other instruments and incorporating additional musical attributes.

Figure 7: Five generated musical measures demonstrating the model's ability to maintain musical coherence and expressive details at TEMPERATURE 0.7.

A TOKEN ID

Table 3: Token IDs for various accents.

Accent	**Token ID**
BAR	27049
BOS	27050
BARRE_NOTE	27051
BEND_NOTE_1	27052
BEND_NOTE_2	27053
BEND_NOTE_3	27054
BEND_NOTE_4	27055
BEND_NOTE_5	27056
BEND_NOTE_6	27057
BEND_NOTE_7	27058
TREM_BAR_1	27059
TREM_BAR_2	27060
TREM_BAR_3	27061
TREM_BAR_4	27062
TREM_BAR_5	27063
DEAD_NOTE	27064
SLIDE_NOTE_1	27065
SLIDE_NOTE_2	27066
SLIDE_NOTE_3	27068
SLIDE_NOTE_4	27069
SLIDE_NOTE_5	27070
SLIDE_NOTE_6	27071
HAMMER	27072
VIBRATO	27073
HARMONIC_1	27074
EOS	27075

B HYPER-PARAMETER TUNING

The following hyper-parameter values were used for training the model: The values were obtained

Table 4: Hyper-parameter values used for training.

Property	**Value**
FFN_HIDDEN	1024
MAX_SEQ_LENGTH	32
NUM_HEADS	4
DROP_PROB	0.3
NUM_LAYERS	2
D_MODEL	256
TEMPERATURE	1.0
OVERLAP	8
LAMBDA	1.5
ALPHA	0.1
DELTA	1.0

after performing a grid search for 10 epochs and selecting values which gave the lowest validation loss. The ranges for the grid search are given in the repository `https://github.com/DjentleViBe/SC0RE`.

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